WEEK 9 · LECTURE L2 OF 3 · UNIT III

Fairness, Transparency and Bias

How bias enters models, why fairness definitions conflict, and what the DPDP Act does and does not do.

Part 1 How bias enters models: proxies and feedback loops

Part 2 Competing fairness definitions; explainability versus accuracy

Part 3 The DPDP Act and employee data

Read Leicht-Deobald et al. (2019); the DPDP employee-data note

R K Singh · Senior Professor, Department of Commerce · Delhi School of Economics, University of Delhi · DSEH001 · M.Com. PGCF 2025

PART 1 · HOW BIAS ENTERS

Four routes into a model

Route	How it works	At GreenLeaf
Historical labels	The model learns from biased past decisions	Past hires, if mostly male
Proxy variables	A neutral input stands in for a protected one	Career gaps for gender; pin code for caste or class
Thin samples	Some groups are rare in training data	Only 19 of 140 requisitions are in Tier 3
Feedback loops	Output shapes the next round of data	Lower allocation score, fewer orders, fewer ratings, lower score

PART 2 · THREE DEFINITIONS

Fair by which definition?

Definition	What must be equal across groups	Met by the flight-risk model?
Demographic parity	Share flagged	No: 72.6% against 24.6%
Equalized odds	False and true positive rates	No: FPR 70.1% against 22.7%
Predictive parity	Precision	Yes: 25.4% against 25.9%

WHY THEY CONFLICT

Base rates differ (22.8% against 19.6% exits). When they do, no imperfect model can satisfy all three. Choosing one is a value judgment you must defend.

PART 3 · THE DPDP ACT

What the Act does and does not do for automated decisions

Does not	Does
Regulate algorithms as such	Require accurate, complete data for decisions
Give a right to an explanation	Require security safeguards and breach reporting
Let consent block staff profiling (employment is a legitimate use)	Require erasure once the purpose is served, and a grievance route

SO Under DPDP, the audit, the human review and a working grievance route are the real safeguards, not consent.

Going deeper: the theory underneath fairness

The next slides set out the impossibility results, Gender Shades and intersectionality, Rudin on interpretable models, and Leicht-Deobald on integrity: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · KLEINBERG ET AL. (2016); CHOULDECHOVA (2017)

Fairness definitions cannot all hold at once

THE CORE IDEA

When groups have different base rates, a model cannot be calibrated and have equal error rates across groups, unless it is perfect.

HOW IT WORKS

Improving one fairness measure necessarily worsens another.

LIMITS AND CRITIQUE

The result is mathematical; deciding which fairness matters is ethical and legal.

AT GREENLEAF GreenLeaf must state which definition it chose for the flight-risk model, and why.

THEORY IN DEPTH · BUOLAMWINI AND GEBRU (2018)

Bias hides at the intersections

THE CORE IDEA

Gender Shades tested commercial face-analysis systems on gender classification.

HOW IT WORKS

Error rates reached 34.7% for darker-skinned women against 0.8% for lighter-skinned men.

LIMITS AND CRITIQUE

One study of one task, but it changed industry practice and showed why single-axis audits miss harm.

AT GREENLEAF

A model fair by gender and fair by age can still be unfair to young women. Audit the intersections.

THEORY IN DEPTH · RUDIN (2019)

Use interpretable models for high-stakes decisions

THE CORE IDEA

Rather than explaining black boxes after the fact, build models people can understand directly.

HOW IT WORKS

In many tabular problems, simple scoring models match black-box accuracy.

LIMITS AND CRITIQUE

Some tasks, such as images and text, still need complex models.

AT GREENLEAF

For HR decisions on 1,000 staff, a transparent scoring rule is likely as accurate as a black box.

THEORY IN DEPTH · LEICHT-DEOBALD ET AL. (2019)

Algorithms can crowd out personal integrity

THE CORE IDEA

When systems decide, people stop exercising moral judgment and simply comply.

HOW IT WORKS

Three erosions: marginalized sense-making, blind trust in rules, loss of moral imagination.

LIMITS AND CRITIQUE

Remedies include critical data literacy and keeping humans accountable for decisions.

AT GREENLEAF

A store manager who says "the system flagged her" has handed away a judgment that remains his.

TAKEAWAYS

Three things to carry out of this lecture

1

Bias enters through labels, proxies, thin samples and feedback loops.

2

Fairness definitions conflict; choose one and defend it.

3

Under DPDP, audit and redress, not consent, protect staff.

NEXT Accountability and algorithmic management of gig workers.